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Semantic Segmentation in Autonomous Driving using Multi-scale Feature Fusion and Graph Network

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Abstract

Semantic segmentation in urban scenes is an important task in computer vision. However, there are many challenges in urban road scenes, such as category imbalance and complex backgrounds. These problems lead to unclear edge segmentation and fuzzy categorization of occluded objects in existing semantic segmentation methods in urban scenes, which limits the accuracy and robustness in practical applications. In this paper, we propose a model that recursively enhances edge feature representation combined with local spatial context. We address the problem of unclear edge segmentation by introducing Multi-scale Central Difference Convolution (MS-CDC) to fuse multi-scale edge features. The FeedBack Connection (FBC) module based on feature pyramid fuses the multi-scale features while recursively augmenting the original network to make the occluded objects more robust. Meanwhile, we design a Local Feature Extraction (LFE) module to capture pixel-wise relationships by constructing local pixel graph and center pixel graph. It can learn local contextual information to extract finer pixel features. Experimental results on the Cityscapes and Mapillary Vista dataset validate the effectiveness of our proposed model. Our model can achieve new results by 80.67% and 45.5% mIoU on the Cityscapes and Mapillary Vista val set. We open-source our code at <https://github.com/sanmanaa/segmentation-autodriving-graph-centralconv>

Keywords: Semantic Segmentation, Autonomous Driving, Local Context aggregation, Multi-scale Feature Fusion

1 Introduction

Semantic segmentation is a fundamental task in computer vision that aims to assign a category label to each pixel. It plays an important role in the understanding of basic urban scenes in autonomous driving. In urban road scenes, semantic segmentation techniques are used to segment categories of interest in images, such as vehicles, driving areas, traffic signals, etc. However, there are extremely complex visual relationships in urban road scenes, with multiple targets obscuring each other, blurred categories, and blurred edges. This results in the task of visual understanding in urban road scenes is extremely difficult.

Early semantic segmentation methods usually relied on hand-crafted features [1]. However, the performance of these methods was far from satisfactory. In the past few years, with the remarkable development of deep learning [2, 3]. Deep Convolutional Neural Networks (DCNNs) [4] have been successfully applied to semantic segmentation. Fully Convolutional Network (FCN) [5] is a pioneering work to employ DCNNs in semantic segmentation, which replace fully connected layers with convolutional layers in the final stage of a typical CNN architecture to obtain better results.

Although these methods have achieved impressive results, as shown in Fig. 1, in urban road scene there are still two current problems: (I) unclear edge segmentation and (II) blurred classification of occluded objects. For example, some parts of the “pole” (i.e., row (I) of Fig. 1) are lost in the segmentation mask and some regions that are occluded from each other by multiple targets (i.e., row (II) and row (III) of Fig. 1) are recognized as misclassified. We find that unclear edge segmentation is due to the fact that the ordinary convolution is not able to capture sufficient edge feature so that the object boundaries cannot be located accurately. Besides, multiple targets misclassification is due to insufficient spatial details in high level feature maps.

To address these problems, the DeepLabv3+ [6] introduces a spatial pyramid pooling module to better understand the differences between large and small targets. Li [1] et al. model the body and edges of the target object to obtain new body features and residual edge features. U-Net [7] adopts an encoder-decoder architecture to improve the semantic segmentation performance. It adds skip connections between the encoder and decoder, which can recover fine-grained details in the semantic prediction. Feature pyramid networks (FPNs) [8] use the structure of U-Net [7] to make predictions from each level of the feature pyramid. FFS-Net [9] designs a multiscale channel attention module to balance the low-level fine information and high-level semantic features. These methods prove that handle asymmetries in deep and shallow features contribute to the problems, but they do not consider extracting multi-scale edge features and context detail information for feature fusion. This information not only helps to localize the edge locations but also helps to improve the segmentation accuracy of mutually occluded targets.

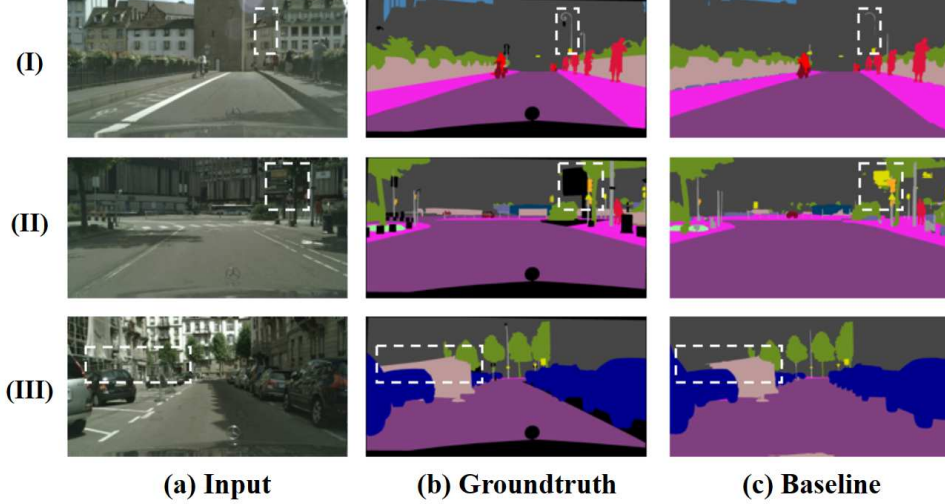


Fig. 1 Two basic problems in our baseline, which are unclear edge segmentation and blurred classification of occluded objects. Samples are from the val set of Cityscapes dataset.

In order to improve the accuracy of segmentation, several pioneering approaches have been proposed to utilize local spatial contextual visual information. Ding [10] et al. introduce a context converter for embedding contextual information in contextual branches and selectively projecting it onto local features. Fang [11] et al. propose a contextual representation enhancement network (CRENet) to enhance global context (GC) and local context (LC) modeling in high-level features. BaGFN [12] addresses implicit cross-feature extraction by proposing an attentive graph fusion network for explicit high-order feature interactions, facilitating context information extraction. Combining local contextual information to re-model the relationships between objects has been shown to be effective in improving the accuracy of semantic segmentation.

In this work, we propose a new model for recursively enhancing feature representation and fusing edge features. The model can extract edge or texture-related features in different perceptual fields on the image, augment the original network recursively to generate increasingly powerful representations. Specifically, to alleviate the existing problem of unclear boundary delineation, the Multi-scale Central Difference Convolution (MS-CDC) module is designed to extract edge texture features of urban roads at multiple scales. We also introduce a pyramid-based FeedBack Connection (FBC) module, which can find occluded targets more easily and enhance the robustness to occluded objects. Thus, the FBC module can alleviate the problem of multiple targets occluding each other. In addition, inspired by methods to improve segmentation accuracy, we design the Local Feature Extraction (LFE) module. The LFE module enhances the feature representation, which mitigates the problem that considering the relationship between all pixels may introduce some irrelevant information.

The main contributions of this paper are summarized as follows:

- Multi-scale Central Difference Convolution (MS-CDC) is designed to extract edge or texture related features in different perceptual fields on an image, which provides

a more robust modeling capability compared to traditional convolution, and thus can improve the problem of unclear boundary segmentation in semantic segmentation tasks.

- The FeedBack Connection (FBC) based on the pyramid feature structure is introduced to add to the original network. It can selectively enhance or inhibit neuronal activation, which effectively enhances the robustness of the occluded object. The method improves the classification ambiguity caused by the mutual occlusion of multiple targets in the segmentation task.

- The Local Feature Extraction (LFE) is designed to capture the pixel-by-pixel relationships in the spatial domain, which learns deep hierarchical features and works in concert to effectively improve the representation of local regions.

2 Related Work

We have extensively surveyed and analyzed recent research advances in the field of semantic segmentation. Next, an in-depth discussion of the remarkable results achieved in recent years in the context of multi-scale fusion and segmentation is provided from two key perspectives. Furthermore, innovative applications of convolutional operators are explored to underscore the representation of semantic information.

2.1 Semantic segmentation

Semantic segmentation has received a lot research in the last few years [6, 11, 13–19], where the Full Convolutional Network (FCN) [5] is the most representative network and most of the existing excellent methods are based on this framework [15, 19–21]. For semantic segmentation, one of the most important things is to ensure that the output feature maps have sufficient semantic and contextual information, while ensuring high resolution. For this purpose, current methods can be divided into those aim at maintaining high resolution [22–24] and those aim at capturing context [6, 20, 25]. In order to maintain high resolution, a common approach is to fuse low-level feature maps into high-level feature maps before outputting [6, 23], so that the output feature maps can meet the requirements of high resolution.

These methods aim to balance the need for high-resolution output with the requirement for rich semantic and contextual information. However, since the feature maps of different levels of these methods are directly up-sampled at the final fusion. This leads to information loss because the up-sampling methods cannot recover the details and precise information present in the low-level features.

2.2 Multi-scale feature fusion

Multi-scale feature fusion is a popular direction in computer vision. FPN [8] combines top-down paths to sequentially combine features at different scales, PANet [26] adds another bottom-up path on top of FPN, and STDL [27] proposes to exploit cross-scale features through a scale transformation module. HRNet [23] connects in parallel from high to low convolutional streams so that it can maintain a high resolution representation throughout the process. BFMNet [28] uses a parallel architecture and

dilated convolution strategy to perceive multi-scale features. MFFTNet [29] proposes multi-scale strip convolution to refine the information present within both deep and low-level feature sets.

These methods improve the model’s ability to perceive multi-scale targets and complex scenes by integrating multi-scale feature information. However, since multi-scale fusion mainly focuses on the overall scene features, it ignores the local features of individual objects in the occluded region. Therefore, they may have difficulty in accurately segmenting individual objects when multiple objects are occluded.

2.3 Context for Segmentation

Pyramid pooling techniques focus on a fixed square context region, as pooling and scaling are typically used in a symmetric manner. Relational context approaches build context by focusing on the relationships between pixels and are not limited to square regions. Such techniques can construct more suitable contexts for non-square semantic regions, such as long trains or slender lamp posts. OCRNet [30] constructs better contexts by aggregating the representation of context pixels, where the context consists of all pixels. MCRNet [31] captures spatial details for high resolutions and semantic encoding for low resolutions. It effectively solves the image blurring problem caused by downsampling operations. Tian [32] et al. propose a context-aware classifier to use additional contextual hints, decently adapting to different latent distributions. Lin [33] et al. propose an attention pyramid transformer to address issues in patch-based transformers, focusing on semantic preservation and global communication.

The core idea of these methods is to capture richer contextual information by considering the relationships between pixels. However, since they need to consider the relationship between pixels in the whole image, their computational complexity is usually high, which may lead to slower training and inference.

2.4 Convolution operator

Convolution operators are commonly used to extract basic visual features in deep learning frameworks. However, convolution operators compute the kernel response to each local patch of the input feature map and are not good at modeling the boundary information. To address this problem, the local binary convolution (LBC) [34] uses a series of predefined binary filters instead of learnable kernels in convolution. Yu [35] et al. introduce the central difference convolutional network (CDCN) to extract intrinsic deception patterns by aggregating intensity and gradient information. Compared to traditional convolution, CDCN can provide a more robust modeling capability. Chen [36] et al. propose DEConv, which employs Difference Convolution (DC) to integrate both low-frequency and high-frequency information. However, it struggles in capturing boundary details. On the other hand, Tan [37] et al. introduced Semantic Difference Convolution (SDC), which leverages semantic similarity to guide CDC. Nevertheless, it lacks perceptual awareness across different scales.

These methods share a commonality in their pursuit of improving feature extraction through diverse convolution operators, aimed at capturing more comprehensive contextual information. However, they collectively face a common limitation that their

suboptimal ability to accurately capture boundary information. This deficiency in boundary awareness may hinder their overall performance in tasks requiring precise boundary delineation.

3 Methodology

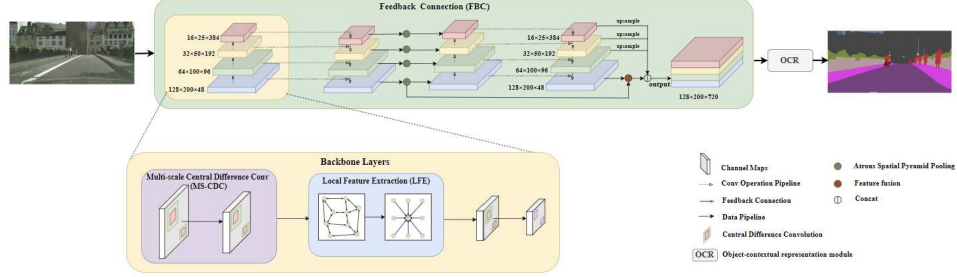


Fig. 2 This paper presents an illustration of a model that fuses multi-scale features and local spatial contexts. It uses LFE module, MS-CDC module and FBC module together to extract more fine-grained semantic features. The model first extracts feature information in each level of resolution using MS-CDC. Second, uses FBC module to perform multi-scale feature fusion. The LFE module using the features extracted by MS-CDC to consider the correlation relationship between pixels and aggregate features in local areas. The OCR (Object-contextual representation) module [30] is used to enhance feature representation based on contextual information

Fig. 2 gives the whole framework for fusing multi-scale features and local spatial context, which consists of three main parts: Multi-scale Central Difference Convolution (MS-CDC) module and FeedBack Connection (FBC) module, Local Feature Extraction (LFE) module. The input image is first sent to the network and then the MS-CDC module is used each time a new low resolution is formed in the main-body to capture the contextual information. After that, we use the proposed LFE module to capture the pixel-by-pixel relationship and aggregate the local features. Thirdly, we incorporate the FBC module when fusing feature maps at different levels. It is used to enhance its robustness in classifying occluded regions. Finally, Hierarchical multi-scale attention [38] outside the backbone network is introduced to improve the accuracy of segmentation.

3.1 Multi-scale Central Difference Convolution (MS-CDC)

For the input images, HRNet [23] produces high-resolution feature representations, achieving state-of-the-art performance in many semantic segmentation datasets. However, 3×3 convolution operation in the shallow layers reduces the image size, which leads to the loss of some fine-grained image features. Therefore, enhancing the perception of fine-grained features at the edges is crucial for semantic segmentation in urban road scenes.

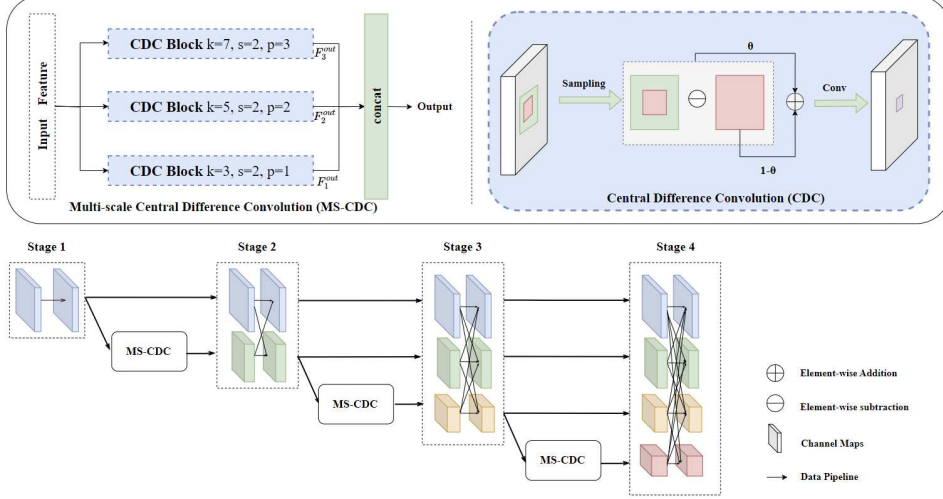


Fig. 3 The left one is the pipeline of the MS-CDC module, in which k ; s ; p denote kernel size, stride and padding. Right one is the architecture of the Central Difference Convolution. The bottom one shows the location of the MS-CDC in the network.

In this paper, we design a new multi-scale central difference convolution module to efficiently extract multi-scale edge features on urban roads. Intensity-level semantic information and gradient-level details are crucial for distinguishing the boundaries of urban roads, which suggests that combining ordinary convolution with center difference is a more feasible way to provide more robust modeling capabilities. Central difference convolution [35] enhances the representation and generalization capabilities of ordinary convolution by aggregating the central gradients of the sampled values. At each resolution level, we use the multi-scale central difference convolution to extract more consistent patterns and biologically motivated features, so that it can capture some fine-grained invariant information. The output feature map y can be formulated as

$$\begin{aligned}
 y(p_0) = & \theta \cdot \sum_{p_n \in \mathcal{R}} \omega(p_n) \cdot (x(p_0 + p_n) - x(p_0)) \\
 & + (1 - \theta) \cdot \sum_{p_n \in \mathcal{R}} \omega(p_n) \cdot x(p_0 + p_n)
 \end{aligned} \tag{1}$$

where p_0 denotes the current position on the input and output feature maps, and p_n exemplifies the position of the local receptive field region $\mathcal{R}$ in the convolution operation; the hyperparameter $\theta \in [0, 1]$ weighs the contribution between the intensity level and the gradient level information. Higher values of θ imply that the importance of the center-differential gradient information is greater. ω 's weights are shared between the convolution and the center-differential convolution, and thus no additional parameters are added.

More robust and biologically motivated features can be captured due to the use of central difference convolutional layers at different scales. As is shown in Fig. 3, we deploy small (3, 3), median (5, 5) and large (7, 7) convolutional kernels in parallel. We concatenate the outputs through feature channels to achieve multiple receptive

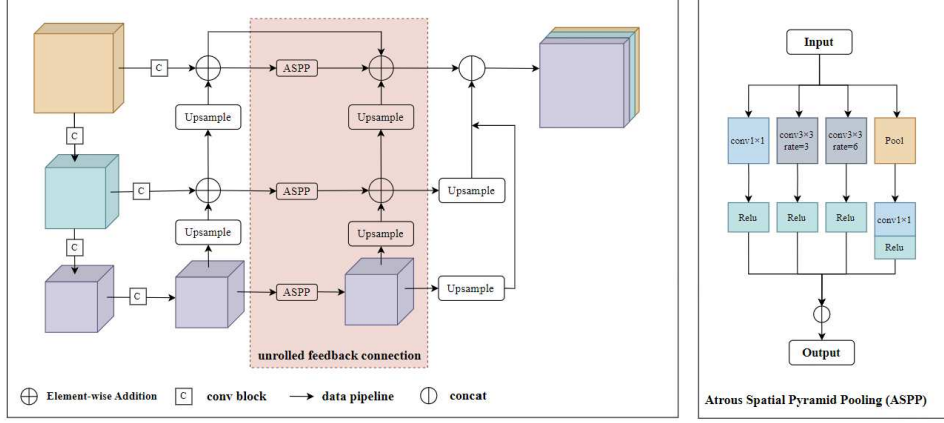


Fig. 4 The feedback connection based on the feature pyramid unfolds as a two-step sequential network, which is shown in the figure for the case $S = 3$. The pink block is the specific structure of unrolled feedback connection.

domains ranging from local to global, so as to achieve effective feature extraction at different scales. The advantages of our designed multi-scale central difference convolution module are twofold. First, it contains different receptive fields to search for texture, edge features at different scales. Second, the central difference convolution layer is able to efficiently extract edge features in the image by aggregating the center gradient.

3.2 FeedBack Connection (FBC)

Although HRNet [23] performs well in handling multi-scale targets with excellent multi-scale feature retention, it is difficult to clearly distinguish between multiple targets when occlusion occurs between them. This lead to imprecision and ambiguity in the segmentation results.

To address this challenge, we introduce FBC module to target the problem in multi-target occlusion scenarios. Human visual perception selectively enhances and suppresses neuron activation, utilizing high-level semantic information through feedback connections. Inspired by this mechanism, computer vision has adopted a similar approach of reevaluating information, akin to "looking and thinking twice". Our Feedback Connections based on feature pyramid operation adds feedback from the bottom-up feature pyramid network to the backbone layer. Let B_i denotes the i th resolution level of the fourth stage of the backbone from low to high, F_i denotes the i th bottom-up feature pyramid layer operation, R_i denotes the feedback connection. The set of feature maps output by the backbone is $\{f_i | i = 1, \dots, S\}$, where S is the total number of resolution levels. The output feature f_i is defined as

$$f_i = F_i(f_{i-1}, x_i), x_i = B_i(x_1, \dots, x_S, R_{i-1}(f_{i-1})) \quad (2)$$

where $R_0 = 0$, $f_0 = 0$, and x_i denotes the features of the input at each resolution level. This makes the feedback connection a recursive operation.

$$f_i^t = F_i^t(f_{i-1}^t, x_i^t), x_i^t = B_i^t(x_1^t, \dots, x_S^t, R_{i-1}^t(f_{i-1}^t)) \quad (3)$$

where $t = 1, \dots, T$, T is the number of iterations of the unfolding and we use the superscript t to denote the operations and features at step T of the unfolding.

We expand the feedback connection into a sequential network with 2 iterations (i.e., $T = 2$). This is shown in Fig. 4. Taking $S = 3$ as an example, we connect f_1, f_2 after two F and we connect the results of the two iterations using Atrous Spatial Pyramid Pooling (ASPP) [14], a module whose input is the feature f_i and transforms it into one of the output features in the feedback connection operation. The module consists of four parallel branches, the outputs of which are then connected together along the channel dimension to form part of the final output of R . Three of the branches in the module use a convolutional layer followed by a ReLU layer, with the number of output channels being one-fourth the number of input channels. The last branch uses a global average pooling layer to compress the features and uses a 1×1 convolutional layer and a ReLU layer to transform the compressed features into one-fourth of the input channel. Finally it is connected to the output features of the other three branches. At this point the output of the ASPP module is the feature connected to the four branches with the same size as the input feature.

3.3 Local Feature Extraction (LFE)

In complex urban scenes, considering relationships between all pixels may introduce irrelevant information, possibly degrading model performance and adding unnecessary complexity. Additionally, neglecting local relationships may hinder capturing vital local details, impacting performance in intricate scenarios. Therefore, LFE module is introduced to effectively extract and utilize local detail information.

Local pixel graph. Given an input feature $F \in \mathbb{R}^{C \times H \times W}$, where C is the number of channels, H is the height, and W is the width, we aim to apply the self-attention mechanism [39] to obtain distinctive local features. First we use three independent 1×1 convolutional layers, an unfold from the torch.nn class in Pytorch, and a rearrange from the einops library to obtain three series of local features $P \in \mathbb{R}^{N \times k^2 \times C}$, $Q \in \mathbb{R}^{N \times C \times k^2}$, $K \in \mathbb{R}^{N \times C \times k^2}$. $N = H \times W$ is the number of pixels and k is the local kernel. In the local pixel graph, the nodes represent all the pixels in the local region, and then the edges represent the relationships between pixels E^{lp} can be defined as

$$E^{lp} = \text{softmax}(P \otimes Q) \quad (4)$$

where $\otimes$ is the matrix multiplication operation and softmax is the softmax function, while we include K to generate more refined features R^{lp} .

$$\begin{aligned} R^{lp} &= E^{lp} \otimes K \\ &= \text{softmax}(P \otimes Q) \otimes K \end{aligned} \quad (5)$$

With E^{lp} and K we obtain the pixel-level refinement of the features R^{lp} . In local pixel graph, we improve the local region at the pixel level by not only connecting pixels in the local space, but also associating pixels with different scales of deflated size [38], an operation that can efficiently augment each local patch by exploiting pixel-level information about the local region.

Central pixel graph. In the central pixel graph, our goal is to aggregate the spatial information in the patch to update the features in the central pixel. This translates patch-level features into pixel-level features. We extract features at the central location based on the shape of the features in the local pixel graph and define an edge as a connection between the center pixel R_c^{lp} , $c = \frac{k^2}{2}$ and all pixels R_j^{lp} , $j \in [1, k^2]$ in the patch. Thus the relation $E^{cp} \in \mathbb{R}^{N \times C \times k^2}$ can be defined as

$$E^{cp} = softmax(R_c^{lp} \odot R_j^{lp}) \quad (6)$$

where $\odot$ denotes element-wise multiplication. Then we could get the refined feature $R^{cp} \in \mathbb{R}^{N \times C}$ of the whole LFE for central pixels of local patches as followings

$$\begin{aligned} R^{cp} &= \sum_{j=1}^{k^2} E^{cp} \odot R_j^{lp} \\ &= \sum_{j=1}^{k^2} softmax(R_c^{lp} \odot R_j^{lp}) \odot R_j^{lp} \end{aligned} \quad (7)$$

In summary, we obtain improved features from the proposed local pixel graph and central pixel graph. First, we refine each local patch by using pixel-level information. Then we obtain the final features by aggregating the local patch features to its central pixel. Compared with traditional non-local modules, our method has less computational overhead and can effectively utilize the local structure information in the image.

4 EXPERIMENTS

4.1 Dataset and Evaluation Metrics

Dataset. We conduct extensive experiments on two popular urban street scene parsing datasets, Cityscapes [40] and Mapillary Vistas [41]. The details of these datasets are given as follows.

Cityscapes [40] is a large-scale urban-scene dataset, holding high-quality pixel-level annotations of 5K images and 20K coarsely annotated images. Finely annotated images consist of 2,975 train images, 500 validation images, and 1,525 test images. The annotations of test images are withheld for benchmarks. The resolution of each image is 2048×1024 , and 19 semantic labels are defined.

Mapillary Vistas research edition [41] comprises 25,000 densely annotated street-level images, distributed across training, validation, and test sets, with counts of

18,000, 2,000, and 5,000 images, respectively. This dataset includes 65 object categories and a void class, featuring image resolutions up to 22 megapixels and varying aspect ratios. It provides a challenging and diverse dataset for semantic segmentation tasks.

In all the experiments on Cityscapes validation set and Mapillary Vistas validation set, we train our models using finely annotated training set for 200K iterations with a total batch size of 2 and a crop size of 1024×1600 in Cityscapes and 512×960 in Mapillary Vista.

Evaluation Metrics. Following[14, 42], the standard Pixel Accuracy (PixAcc) and mean Intersection of Union (mIoU) are used as the evaluation metrics on model effectiveness. Besides, to verify the model efficiency, the model Parameters (Params) is also taken into account.

4.2 Implementation Details

Our model is trained on an NVIDIA RTX A6000 with mixed precision, distributed data parallel training and simultaneous batch normalization.

Backbone. In this paper, we choose the commonly used HRNetV2-W48 [30] as the backbone network for feature extraction for ablation studies and performance comparison with state-of-the-art methods.

Semantic Head. Semantic predictions are performed by a dedicated fully convolutional head consisting of $(3 \times 3conv) \rightarrow (BN) \rightarrow (ReLU) \rightarrow (3 \times 3conv) \rightarrow (BN) \rightarrow (ReLU) \rightarrow (1 \times 1conv)$. The final convolution outputs num_classes channels.

Baseline. Our baseline model is the Hierarchical Multi-scale Attention Model [38]. This model combines multiple scales for prediction using an attention-based approach. It learns dense masks for each scale and combines these multi-scale predictions by performing pixel-by-pixel multiplications between masks. Finally, it performs pixel-wise summation across scales to obtain the final result. In this work, our proposed network follows the same training procedure.

Model details. We use Stochastic Gradient Descent (SGD) as the optimizer. The batchsize of each GPU is 2, momentum 0.9, and the weight decay in training is 5e-4. We apply a "polynomial" learning rate strategy [43]. We use RMI [44] as the primary loss function with default settings and cross-entropy as the secondary loss function. For cityscapes, we use a poly exponent of 2.0, an initial learning rate of 0.05, and train for 175 epochs. As in [38], we use class uniform sampling in the data loader to sample each class equally, which helps to improve the results when the data is unevenly distributed.

For data augmentation, we employ gaussian blur, color augmentation, random horizontal flip and random scaling (0.5x-2.0x) on the input images to augment the dataset the training process.

4.3 Comparisons with the State-of-the-art Methods

In this section, we make result comparisons between our proposed model and the state-of-the-art approaches quantitatively and qualitatively.

Table 1 Quantitative benchmarks of applying the proposed method to some segmentation methods with different backbone networks, while testing on the whole image. SI: sidewalk; BU: building; TL: traffic light; TS: traffic sign; VE: vegetation; TE: terrain; PE: person; MO: motorcycle; BI: bicycle.

Backbone	Method	Road	SI	BU	Wall	Fence	Pole	TL	TS	VE	TE	Sky	PE	Rider	Car	Truck	Bus	Train	MO	BI	mIoU
HRNetV2-W48[23]	HMS[38]	97.8	85.1	94.4	48.4	63.8	69.9	72.0	80.4	91.8	52.8	95.6	88.3	71.7	95.5	71.4	76.0	86.8	76.2	79.9	78.8
HRNetV2-W48[23]	+Ours	98.0	86.6	94.8	47.4	64.3	71.7	72.8	80.9	92.2	58.6	95.5	88.8	73.3	95.8	79.5	83.3	90.1	78.5	80.1	80.7
ResNet-101[45]	HANet[46]	98.3	86.2	92.1	62.3	65.1	62.0	61.8	74.1	92.0	64.8	94.7	77.2	57.4	94.3	79.1	88.4	81.3	51.7	73.0	76.6
ResNet-101[45]	+Ours	98.3	86.6	92.1	63.7	69.3	63.7	69.1	78.8	92.1	65.4	94.4	80.3	65.3	94.8	79.7	84.2	77.9	61.3	76.5	78.6

Table 2 Comparison with the state-of-the-art methods on the Cityscapes validation set.

Method	Backbone	mIoU
PSPNet[19]	ResNet-101	78.4
FCN[5]	ResNet-101	75.1
EncNet[25]	ResNet-101	78.6
Deeplabv3+[22]	ResNet-101	79.3
ANN[47]	ResNet-101	77.1
MaskFormer[48]	ResNet-101	80.3
HANet[46]	ResNet-101	76.6
HMS[38]	HRNetV2-W48	78.8
HRNet[23]	HRNetV2-W48	79.5
Segmenter[49]	ViT-L	79.1
StructToken[50]	ViT-L	80.1
Ours	HRNetV2-W48	80.7

Quantitative results on Cityscapes [40]. The quantitative benchmarks of our proposed model application with different segmentation trunks are shown in TABLE 1. Specifically, the proposed method achieves in average 2.0% mIoU improvement on ResNet-101 backbone, and 1.85% mIoU improvement on HRNetv2-w48 backbone. It is noteworthy that the proposed model consistently improves the baseline on categories of small-size objects (e.g., traffic light, person, and rider) and with object that can be easily obscured (e.g., terrain, fence). These quantitative results show that our model is effective. The key reason is the adept integration of the MS-CDC module, FBC module, and LFE module in our proposed model. They together effectively improve the accuracy of semantic segmentation.

To demonstrate the effectiveness of the proposed model, we compare it against the existing state-of-the-art semantic segmentation approaches on the val set of Cityscapes. TABLE 2 shows results of our model and other methods [5, 19, 22, 23, 25, 38, 46, 49] on the Cityscapes val set. Compared to some methods using ResNet-101 as the backbone, such as MaskFormer [48] and HANet [46]. Our model achieves in 0.3%-5.5% mIoU improvement, affirming the efficiency of our proposed model in semantic segmentation. Besides, our model is superior to HMS [38] and HRNet[23], achieving an mIoU of 1.9% and 1.2-%. It demonstrates that we can effectively leverages the multi-resolution features of the HRNetV2-W48 backbone, allowing for better capturing of object details and contextual information. Meanwhile, the performance

of our proposed model is better than Vision Transformer Large (ViT-L) based models, exemplified by Segmenter [49] and StructToken [50]. It means our model indicates the superiority over the Transformer architecture. Overall, the results of Cityscapes further indicate our proposed model can improve segmentation quality by alleviating the problems of unclear edge segmentation and blurred classification of occluded objects.

Table 3 Comparison with the state-of-the-art methods on the Mapillary Vista validation set.

Method	Backbone	mIoU
Deeplabv3+[22]	ResNet-101	37.4
RGPNet[51]	ResNet-101	41.7
ADANet[52]	ResNet-50	42.5
MaskFormer[48]	Swin-S	42.2
OCRNet [30]	HRNetV2-W48	38.3
FPMN[53]	ResNet-50	44.1
HSSN[54]	Swin-S	44.0
Ours	HRNetV2-W48	45.5

Quantitative results on Mapillary Vista [41]. In this section, we conduct experiments on the much challenging Mapillary Vista [41] dataset. Table 3 shows results of our approach and other methods[22, 30, 48, 51–54] on the val set. Our method achieves 45.5% mIoU for segmentation performance. We outperform the other existing approaches, and especially surpass the Deeplabv3+[22] with 8.1%. Meanwhile, the performance of our proposed model is better than Swin-S based models, exemplified by MaskFormer[48] and HSSN[54]. These results clearly demonstrate the efficacy of our semantic segmentation framework.

Qualitative Results on Cityscapes [40]. Fig. 5 shows some visualization results of the baseline and our model on the val set of Cityscapes [40]. Compared to the HMS [38] results, we can observe that our model can segment the objects more accurately. Concretely, we can well recognize edge regions of fine object classes, such as the cases of the first three rows in Fig. 5. We can underline that our model tries to segment edge of “pole” in the white box. This is because we extract edge features at different scales of the receptive field, introducing enough detail in the edge texture. Furthermore, we can observe that our model can alleviate the problem of multiple objects occluding each other, as far as the last three rows in Fig. 5, the baseline can not classify the “fence”, “pole” and “traffic lights” correctly. On the contrary, our model not only corrects the mistake, but also segment correctly. These results show that our proposed model can alleviate the problem of edge blurring with multiple objects occluding each other. In summary, the above visualization results can visually verify the effectiveness of our method.

Qualitative Results on Mapillary Vista [41]. Fig. 6 presents visual comparisons between our proposed method and the baseline. Despite the relatively low mIoU values, our segmentation results for Mapillary dataset are evident, effectively delineating narrow and large objects like traffic lights, cars, and roads on the streets.

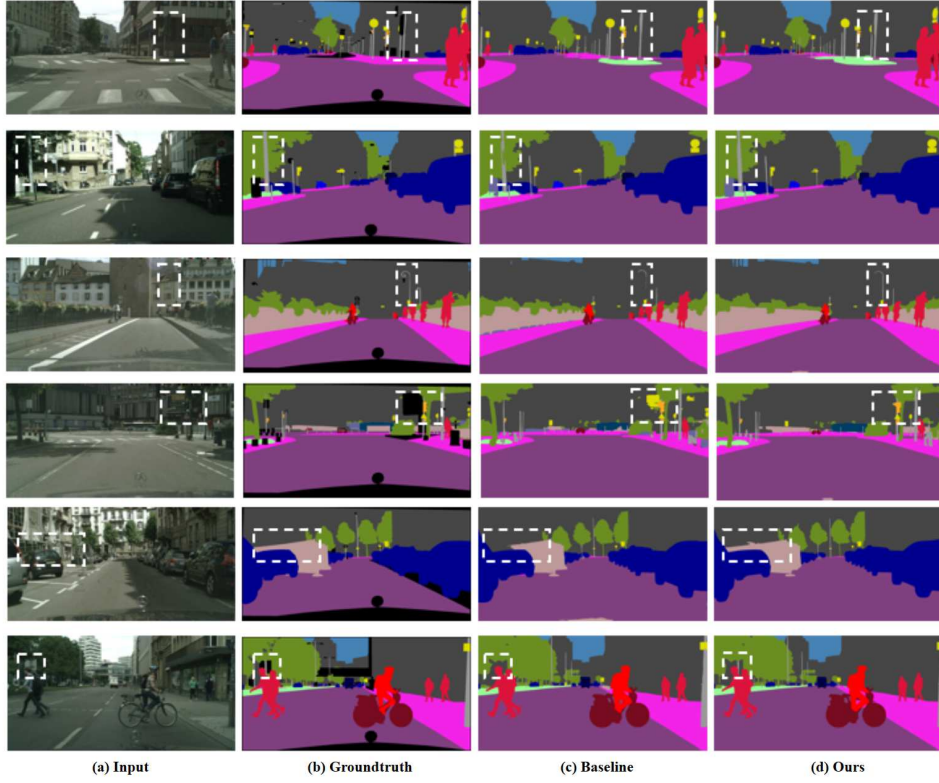


Fig. 5 Qualitative visualization of input images, ground truth, our baseline and ours. The examples are chosen from Cityscapes dataset. Compared to the baseline, our model reduces unclear edge segmentation and multiple objects occlude errors as highlighted in white box.



Fig. 6 Qualitative visualization of input images, ground truth, our baseline and ours. The examples are chosen from Mapillary Vista dataset. Compared to the baseline, our model reduces unclear edge segmentation and multiple objects occlude errors as highlighted in red box.

Table 4 ABLATION STUDIES ON THE VAL SET OF CITYSCAPES [40].

Baseline	MS-CDC	FBC	LFE	mIoU(%)	PixAcc(%)	Params(M)
✓				78.813	96.323	72.1
✓	✓			79.442	96.386	74.3
✓		✓		79.514	96.533	79.9
✓			✓	80.344	96.588	72.3
✓	✓	✓		80.613	96.642	82.1
✓	✓	✓	✓	80.670	96.596	82.2

Specifically, our method effectively detects object edges, such as the road signs in the first row of Fig. 6 and poles in the third row. This improvement is attributed to the integration of the MS-CDC module for finer edge segmentation. Furthermore, we observe an enhancement in addressing the issue of blurred classification due to occlusion of multiple targets, as depicted in the second row of Fig. 6. This indicates our method’s capability to capture local contextual information.

4.4 Ablation Study

Our ablation study aims to investigate the effectiveness of the Multi-scale Center Difference Convolution (MS-CDC) module, the feature pyramid-based FeedBack Connection (FBC) module, and the Local Feature Extraction (LFE) module and their combinations in segmentation. For this purpose, we conducted a series of experiments on the val set of cityscapes. TABLE 4 shows the experimental results compared to the baseline. The experimental results are listed with mIoU, PicAcc and model parameters for comparison.

Effectiveness of MS-CDC. As shown in the first and second rows of TABLE 4, we can observe that with the help of the MS-CDC, the model can achieve the mIoU by 79.442% and PixAcc by 96.386%. Compared to the baseline model [38], MS-CDC has a performance gain of 0.629% mIoU. These results empirically illustrate that using central difference convolution instead of traditional convolution as well as aggregating multi-scale edge features play an important role in the segmentation of urban road scenes.

Effectiveness of FBC. From the first and third rows of TABLE 4, compared to results of the baseline model [38], we can observe that FBC has the performance gain by mIoU of 0.7% and PixAcc of 0.2%, which not only verifies the importance of merging shallow feature maps into deep ones for segmentation task, but also verifies the necessity of adding recursive feedback connection to the backbone network.

Effectiveness of LFE. Based on the first and fourth rows of TABLE 4 and comparing it with the baseline model’s results [38], we can see that LFE demonstrates improved performance. It shows an increase in mIoU by 1.5% and PixAcc by 0.26%. This not only highlights the importance of enhancing feature representation in local regions but also confirms the effectiveness of the local graph network in segmenting urban scenes.

Effectiveness of Module Efficiency. From the perspective of model parameters, As shown in TABLE 4, the baseline model uses the least model parameters by 72.1

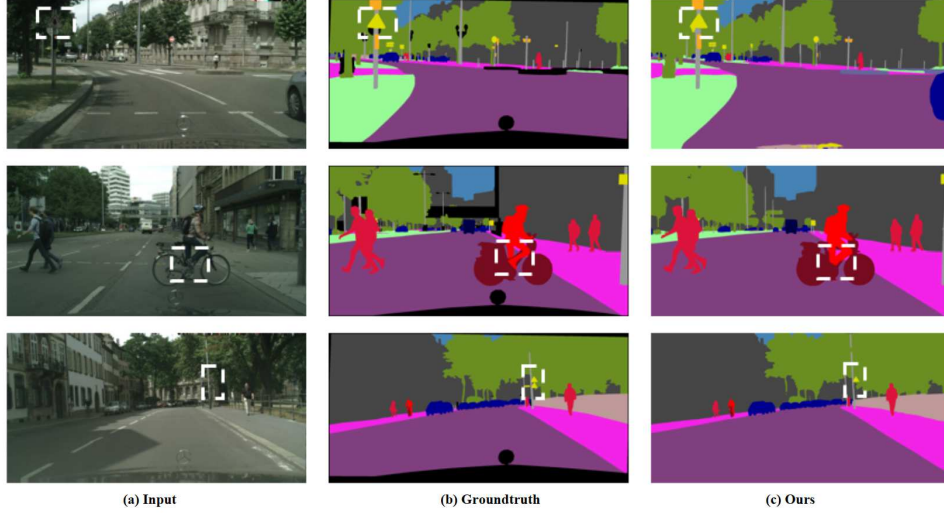


Fig. 7 The failure cases on the val set of Cityscapes. The white dashed frames highlight the failed areas predicted by our model.

M, while our FBC has more computation parameters and slightly surpasses it by 7.8 M. This is because our FBC can selectively enhance or inhibit neuronal activation and enable better multi-scale information fusion. Despite the parameter increase, we contend that the performance enhancement justifies this adjustment. The experimental results confirm that our FBC can somewhat alleviate the occluded object problem without introducing too many model parameters. Meanwhile, our MS-CDC module and LFE module achieve higher efficiency, requiring only 2.2M and 0.2M parameters respectively to effectively extract edge features and correlate local information. They address the issue of unclear edge segmentation and enhance segmentation accuracy.

4.5 Shortcomings and Areas for Improvement

Our model excels in semantic segmentation tasks, particularly in capturing object details and contextual information. However, it is important to note that our model also has its limitations. Failure cases are shown in the Fig. 7. We can observe that there are still imperfections in some special kind of object segmentation masks. For example, the cases of the first and third rows in Fig. 7, the traffic sign is hard to be accurately segmented due to the insufficient resolution representation from the backbone network. In addition, some background stuff usually appears with the foreground objects together like the "rider" and "bicycle". They are essentially a kind of spurious associations that background pixels can easily be confused with objects, since these kinds of objects often appear together in the used dataset. This problem can be served by removing spurious associations between different categories of objects through the causal intervention strategy.

Furthermore, in addressing the challenges within object segmentation, we can explore the use of deep learning techniques and reinforcement learning methods to

enhance the model’s ability to identify and segment specific object categories. This may involve the introduction of advanced attention mechanisms to enhance the model’s focus on key object features and its self-correction capabilities. Additionally, considering the incorporation of additional modalities, such as depth information or time-series data, may improve the accuracy of object segmentation. Ultimately, the integration of a variety of technologies and strategies can contribute to better tackling the challenges in object segmentation.

5 CONCLUSION

We introduce a novel network for semantic segmentation that achieves multi-scale feature fusion, edge feature extraction and aggregation of local feature information through three key schemes (MS-CDC, FBC, and LFE). The model collectively address challenges such as unclear edge segmentation and blurred classification of occluded objects in urban scenes. Experimental results on the challenging Cityscapes dataset and Mapillary Vista dataset validate the effectiveness and efficiency of this network.

Our model demonstrate remarkable performance in semantic segmentation. However, we also come with challenges, such as potential increases in computational complexity due to increased resolution and the co-occurrent problem between the background and the foreground. Nevertheless, the network shows promise in handling multi-scale, occlusion, and detail issues in urban environments. Future work could focus on further optimizing computational efficiency, introducing some extra-data multi-modal (e.g., sentences, and extra-images) into semantic segmentation and enhancing the network’s capability to handle more complex scenarios, including dynamic objects, diverse lighting conditions, and environmental factors.

6 Data and code availability

The datasets and code supporting the findings of this study are available as follows:

- **Nature of the data:** The data used in this study consists of annotated images for semantic segmentation in autonomous driving, sourced from the Cityscapes and Mapillary datasets. The code includes scripts for data preprocessing, model training, and evaluation.
- **Access to the data:**
 - **Cityscapes**[40]: Available at <https://www.cityscapes-dataset.com>.
 - **Mapillary Vistas**[41]: Available at <https://www.mapillary.com/dataset/vistas>.
- **Access to the code:**
 - **Zenodo:** <https://zenodo.org/doi/10.5281/zenodo.12539720>
- **Restrictions on data access:** There are no restrictions on access to the code, which is available under an open-access license. The Cityscapes and Mapillary Vistas datasets are also openly accessible, subject to their respective terms of use.

The accession number for the code is included in the provided DOI. For any materials that must be obtained through a Material Transfer Agreement (MTA), please contact the corresponding author for further details.

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